3. **Forgetting the $1/T$ scaling.** Replicas are scaled by $1/T$; the reconstruction filter compensates with gain T.
4. **Thinking aliasing can be undone.** It cannot. Prevent it before sampling.
5. **Claiming non-band-limited signals can't be sampled.** In practice, we band-limit with an anti-aliasing filter first.

12 Practice Questions

Test your understanding with these quick questions. Answers are at the bottom of the section.

1. A signal $x(t)$ has $X(j\omega) = 0$ for $|\omega| > 5000\pi$. What is the Nyquist rate? What is the maximum allowable sampling period T?
2. True or false: if $\omega_s = 2\omega_M$, the sampling theorem guarantees perfect reconstruction.
3. In one sentence, explain *why* sampling in time produces copies of the spectrum in the frequency domain.
4. A signal is sampled and the resulting spectrum $X_p(j\omega)$ shows overlapping replicas. Can you undo the overlap with a clever filter? Why or why not?
5. What is an anti-aliasing filter, and *where* in the signal chain does it go (before or after the sampler)?
6. Determine the Nyquist rate for $x(t) = \cos(600\pi t) + \cos(1800\pi t)$.
7. A signal with $\omega_M = 3000\pi$ is sampled with $T = 2 \times 10^{-4}\,\text{s}$. Compute ω_s and state whether aliasing occurs.
8. You sample $x(t) = \cos(2\pi \cdot 900\,t)$ at $f_s = 1000\,\text{Hz}$. What frequency does the reconstructed signal have?
9. Repeat the previous question for $f_s = 1500\,\text{Hz}$.
10. A signal has Nyquist rate ω_0. What is the Nyquist rate of $x(t-5)$? What about $x(2t)$?
11. The ideal reconstruction filter has gain T in its passband. Why is that factor of T needed? (Hint: look at the $1/T$ scaling of the replicas.)
12. A zero-order hold is used instead of an ideal lowpass filter for reconstruction. Is the result exact? What kind of error does it introduce?
13. A movie camera shoots at 24 frames per second. A helicopter blade spins at 25 revolutions per second. Describe qualitatively what the blade looks like in the film.
14. You are told that $X(j\omega) = 0$ for $|\omega| > 10{,}000\pi$. Someone samples $x(t)$ at $f_s = 8000\,\text{Hz}$. Is the sampling theorem satisfied? If not, what is the minimum f_s that works?
15. Suppose $x(t)$ is *not* band-limited (its spectrum extends to $\pm\infty$). Does this mean we can never sample it in practice? Explain what is done in real systems.

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